CPU Scheduling: Student Assignment Worksheet

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Instructions for all problems

1. Draw a Gantt chart, including any CPU idle periods.
2. Find each process's completion time (CT).
3. Calculate each process's turnaround time (TAT) and waiting time (WT).
4. Calculate average turnaround time and average waiting time.

Formulas

TAT = CT − AT

WT = TAT − BT

Average TAT = Sum of TAT values ÷ Number of processes

Average WT = Sum of WT values ÷ Number of processes

Assumptions and tie-breaking rules

- All times are in milliseconds (ms). There is one CPU, no I/O waiting, and zero context-switch overhead.
- If processes arrive simultaneously, use their listed order.
- For equal burst times, remaining times, or priorities, prefer the earlier-arriving process, then the listed order.
- Do not preempt a running process merely because another process has equal remaining time or equal priority.
- Smaller priority numbers mean higher priority.
- In Round Robin, arrivals at the exact end of a time quantum enter the queue before the unfinished running process is requeued.

Problem 1: FCFS — All Processes Arrive Together

Schedule the following processes using First Come, First Served.

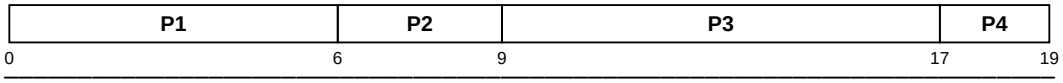
Process	Arrival time (ms)	Burst time (ms)
P1	0	6
P2	0	3
P3	0	8
P4	0	2

Complete all tasks from the general instructions, then answer the additional questions.

Additional questions

- Which process has the longest waiting time?

Gantt chart



Calculations

Process	CT (ms)	TAT (ms)	WT (ms)
P1	6	6	0
P2	9	9	6
P3	17	17	9
P4	19	19	17

Average TAT: **12.75** ms Average WT: **8** ms

Additional answers / working:

Longest waiting time: P4 (17 ms).

FCFS order: P1, P2, P3, P4. WT = 0, 6, 9, 17 (sum 32, avg 32/4 = 8). TAT = 6, 9, 17, 19 (sum 51, avg 51/4 = 12.75).

Problem 2: FCFS — Different Arrival Times and CPU Idle Time

Schedule the following processes using FCFS.

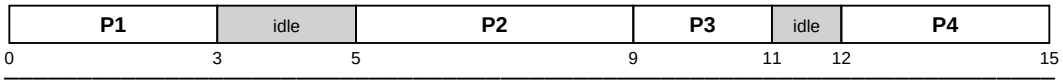
Process	Arrival time (ms)	Burst time (ms)
P1	0	3
P2	5	4
P3	6	2
P4	12	3

Complete all tasks from the general instructions, then answer the additional questions.

Additional questions

- Mark every CPU idle interval.
- Calculate the total idle time between time 0 and the completion of all processes.

Gantt chart



Calculations

Process	CT (ms)	TAT (ms)	WT (ms)
P1	3	3	0
P2	9	4	0
P3	11	5	3
P4	15	3	0

Average TAT: **3.75** ms Average WT: **0.75** ms

Additional answers / working:

Idle intervals: 3 to 5 ms (P2 has not arrived) and 11 to 12 ms (P4 has not arrived).

Total idle time = 2 + 1 = 3 ms.

TAT sum = 3 + 4 + 5 + 3 = 15, avg 3.75. WT sum = 0 + 0 + 3 + 0 = 3, avg 0.75.

Problem 3: SJF — All Processes Arrive Together

Schedule the following processes using non-preemptive Shortest Job First.

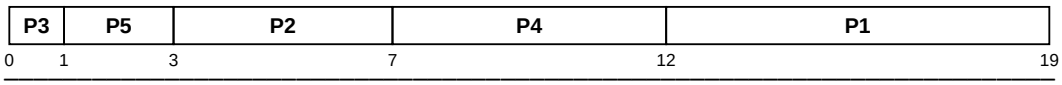
Process	Arrival time (ms)	Burst time (ms)
P1	0	7
P2	0	4
P3	0	1
P4	0	5
P5	0	2

Complete all tasks from the general instructions, then answer the additional questions.

Additional questions

- Repeat the scheduling using FCFS. Which algorithm gives a lower average waiting time for this workload?

Gantt chart



Calculations

Process	CT (ms)	TAT (ms)	WT (ms)
P1	19	19	12
P2	7	7	3
P3	1	1	0
P4	12	12	7
P5	3	3	1

Average TAT: _____**8.4**_____ ms Average WT: _____**4.6**_____ ms

Additional answers / working:

SJF order (shortest burst first): P3(1), P5(2), P2(4), P4(5), P1(7).

FCFS (P1, P2, P3, P4, P5): CT = 7, 11, 12, 17, 19. WT = 0, 7, 11, 12, 17. Avg WT = 47/5 = 9.4 ms (Avg TAT = 66/5 = 13.2 ms).

SJF avg WT = 4.6 ms, which is lower than FCFS (9.4 ms). So SJF gives the lower average waiting time.

Problem 4: SJF — Different Arrival Times

Schedule the following processes using non-preemptive SJF.

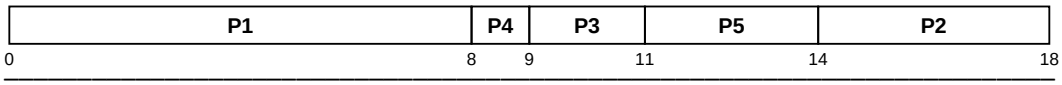
Process	Arrival time (ms)	Burst time (ms)
P1	0	8
P2	1	4
P3	2	2
P4	3	1
P5	10	3

Complete all tasks from the general instructions, then answer the additional questions.

Additional questions

- Which process runs immediately after P1?
- Why does P4 not interrupt P1 when it arrives?

Gantt chart



Calculations

Process	CT (ms)	TAT (ms)	WT (ms)
P1	8	8	0
P2	18	17	13
P3	11	9	7
P4	9	6	5
P5	14	4	1

Average TAT: **8.8** ms Average WT: **5.2** ms

Additional answers / working:

Process that runs immediately after P1: P4. At t = 8, P2, P3 and P4 are waiting, and P4 has the shortest burst (1 ms).

P4 does not interrupt P1 because non-preemptive SJF never takes the CPU from a running process. P1 keeps the CPU until it finishes at t = 8.

TAT sum = 44, avg 8.8. WT sum = 26, avg 5.2.

Problem 5: SRTF — Preemptive Scheduling

Schedule the following processes using Shortest Remaining Time First.

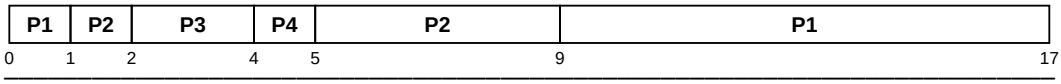
Process	Arrival time (ms)	Burst time (ms)
P1	0	9
P2	1	5
P3	2	2
P4	4	1

Complete all tasks from the general instructions, then answer the additional questions.

Additional questions

- Identify every preemption: a running process being interrupted before completing.
- At time 4, which process should run next, and why?

Gantt chart



Calculations

Process	CT (ms)	TAT (ms)	WT (ms)
P1	17	17	8
P2	9	8	3
P3	4	2	0
P4	5	1	0

Average TAT: **7** ms Average WT: **2.75** ms

Additional answers / working:

Preemptions: (1) at t = 1, P2 (burst 5) preempts P1 (remaining 8). (2) at t = 2, P3 (burst 2) preempts P2 (remaining 4).

At time 4, P4 runs next. P3 has just finished and P4 has the smallest remaining time (1), against 4 for P2 and 8 for P1.

TAT sum = 28, avg 7. WT sum = 11, avg 2.75.

Problem 6: Round Robin — Time Quantum of 2 ms

Schedule the following processes using Round Robin with a time quantum of 2 ms.

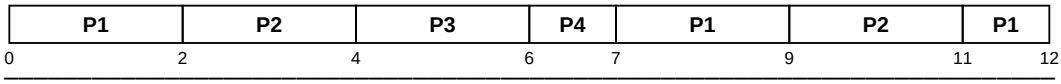
Process	Arrival time (ms)	Burst time (ms)
P1	0	5
P2	0	4
P3	0	2
P4	0	1

Complete all tasks from the general instructions, then answer the additional questions.

Additional questions

- Write the ready-queue order immediately after each time slice.
- Which process finishes first?

Gantt chart



Calculations

Process	CT (ms)	TAT (ms)	WT (ms)
P1	12	12	7
P2	11	11	7
P3	6	6	4
P4	7	7	6

Average TAT: **9** ms Average WT: **6** ms

Additional answers / working:

Ready queue after each slice (front to back): t=2: P2, P3, P4, P1 | t=4: P3, P4, P1, P2 | t=6: P4, P1, P2 | t=7: P1, P2 | t=9: P2, P1 | t=11: P1 | t=12: empty.

First process to finish: P3 (at t = 6).

TAT sum = 36, avg 9. WT sum = 24, avg 6.

Problem 7: Round Robin — Processes Arrive During Execution

Schedule the following processes using Round Robin with a time quantum of 3 ms.

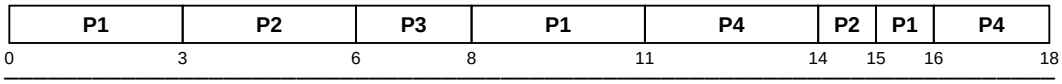
Process	Arrival time (ms)	Burst time (ms)
P1	0	7
P2	1	4
P3	3	2
P4	5	5

Complete all tasks from the general instructions, then answer the additional questions.

Additional questions

- Show the ready-queue order immediately after P1's first time slice.
- Explain where P3 is placed when it arrives at time 3. Apply the boundary-arrival rule.

Gantt chart



Calculations

Process	CT (ms)	TAT (ms)	WT (ms)
P1	16	16	9
P2	15	14	10
P3	8	5	3
P4	18	13	8

Average TAT: **12** ms Average WT: **7.5** ms

Additional answers / working:

Ready queue immediately after P1's first slice (t = 3), front to back: P2, P3, P1.

P3 arrives at exactly t = 3, the end of P1's quantum. By the boundary-arrival rule, the arriving process joins the queue before the unfinished running process is requeued, so P3 is placed ahead of P1.

TAT sum = 48, avg 12. WT sum = 30, avg 7.5.

Problem 8: Non-Preemptive Priority Scheduling

Schedule the following processes using non-preemptive priority scheduling.

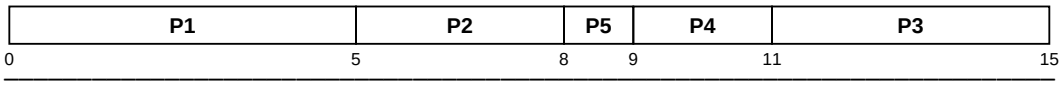
Process	Arrival time (ms)	Burst time (ms)	Priority
P1	0	5	3
P2	1	3	1
P3	2	4	4
P4	3	2	2
P5	4	1	1

Complete all tasks from the general instructions, then answer the additional questions.

Additional questions

- Why does P1 continue running after P2 arrives?
- When the CPU becomes free, how is the tie between P2 and P5 resolved?

Gantt chart



Calculations

Process	CT (ms)	TAT (ms)	WT (ms)
P1	5	5	0
P2	8	7	4
P3	15	13	9
P4	11	8	6
P5	9	5	4

Average TAT: **7.6** ms Average WT: **4.6** ms

Additional answers / working:

P1 continues after P2 arrives because the algorithm is non-preemptive: a higher-priority arrival cannot take the CPU from a running process.

When the CPU becomes free at t = 5, P2 and P5 both have priority 1. P2 is chosen because it arrived earlier (t = 1, against t = 4).

TAT sum = 38, avg 7.6. WT sum = 23, avg 4.6.

Problem 9: Preemptive Priority Scheduling

Schedule the following processes using preemptive priority scheduling.

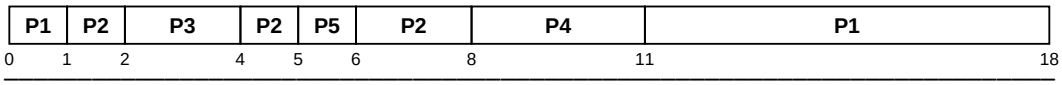
Process	Arrival time (ms)	Burst time (ms)	Priority
P1	0	8	4
P2	1	4	2
P3	2	2	1
P4	3	3	3
P5	5	1	1

Complete all tasks from the general instructions, then answer the additional questions.

Additional questions

- Identify every preemption.
- Which process has the longest waiting time?

Gantt chart



Calculations

Process	CT (ms)	TAT (ms)	WT (ms)
P1	18	18	10
P2	8	7	3
P3	4	2	0
P4	11	8	5
P5	6	1	0

Average TAT: **7.2** ms Average WT: **3.6** ms

Additional answers / working:

Preemptions: (1) t = 1: P2 (priority 2) preempts P1 (priority 4). (2) t = 2: P3 (priority 1) preempts P2. (3) t = 5: P5 (priority 1) preempts P2.

Longest waiting time: P1 (10 ms).

TAT sum = 36, avg 7.2. WT sum = 18, avg 3.6.

Problem 10: Compare Four Scheduling Algorithms

Solve using FCFS, non-preemptive SJF, SRTF, and Round Robin with a time quantum of 2 ms.

Process	Arrival time (ms)	Burst time (ms)
P1	0	6
P2	1	3
P3	2	1
P4	3	4

Complete all tasks from the general instructions, then answer the additional questions.

Additional questions

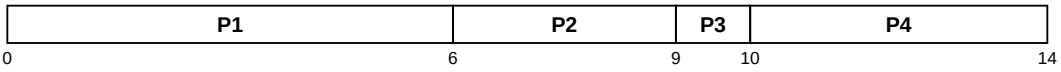
- Which algorithm gives the lowest average waiting time for this workload?
- Does that algorithm also give the lowest average turnaround time?
- Do any algorithms produce the same execution order for this particular process set?

Algorithm	Average TAT (ms)	Average WT (ms)
FCFS	8.25	4.75
SJF	7.75	4.25
SRTF	6.25	2.75
Round Robin (q = 2 ms)	8.75	5.25

Working space

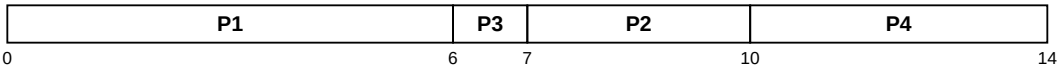
Use an additional sheet for the four Gantt charts and process calculations.

FCFS:



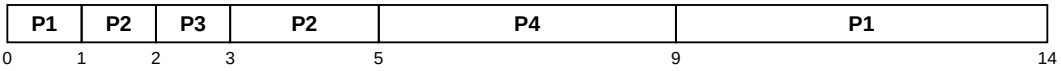
CT: P1=6, P2=9, P3=10, P4=14 | TAT: 6, 8, 8, 11 (avg 8.25) | WT: 0, 5, 7, 7 (avg 4.75)

SJF (non-preemptive):



CT: P1=6, P2=10, P3=7, P4=14 | TAT: 6, 9, 5, 11 (avg 7.75) | WT: 0, 6, 4, 7 (avg 4.25)

SRTF:



CT: P1=14, P2=5, P3=3, P4=9 | TAT: 14, 4, 1, 6 (avg 6.25) | WT: 8, 1, 0, 2 (avg 2.75)

Round Robin (q = 2 ms):



CT: P1=12, P2=10, P3=5, P4=14 | TAT: 12, 9, 3, 11 (avg 8.75) | WT: 6, 6, 2, 7 (avg 5.25)

Answers to the additional questions:

Lowest average waiting time: SRTF (2.75 ms).
Does SRTF also give the lowest average turnaround time? Yes (6.25 ms), because TAT = WT + BT and the burst times are the same for every algorithm.
Same execution order? No. All four algorithms give different execution orders for this process set. FCFS and SJF are the closest, but SJF runs P3 before P2.